

THE SCOTS COLLEGE



YEAR 12 UNIT MATHEMATICS HSC TRIAL EXAMINATION

AUGUST 2011

INSTRUCTIONS TO CANDIDATES:

- Reading time: 5 minutes
- Working time: 3 hours
- Write using blue or black pen (sketches can be in pencil)
- Board approved calculators may be used
- All 10 questions are of equal value(12 marks each).
- Answer each question in a separate booklet
- All necessary working should be shown in every question
- A table of standard integrals is provided
- Total marks: 120
- Weighting: 40%

Students are advised that this is a trial examination only and cannot in any way guarantee the content or format of the Higher School Certificate examination.

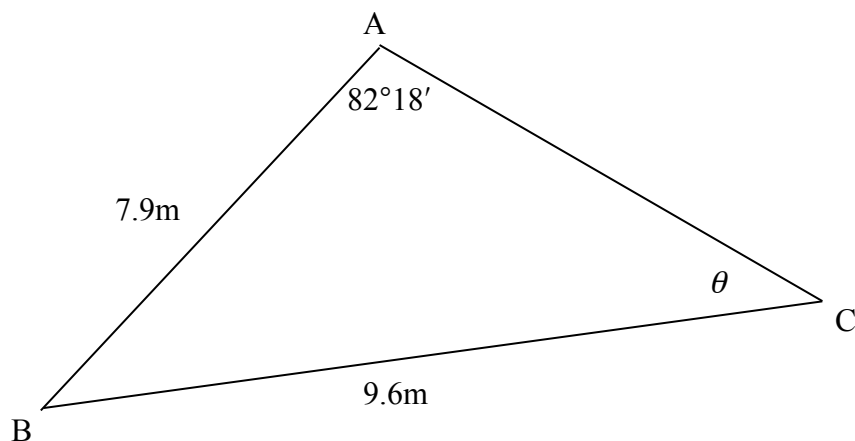
Question 1(Start a New Booklet) (12 Marks)

- (a) Find the reciprocal of $3\frac{4}{7}$ 1
- (b) Evaluate $e^{-0.6}$ correct to three decimal places. 1
- (c) Solve $4 - 5x \leq 3$ 2
- (d) Evaluate $\int_0^2 3e^{2x} dx$ (Answer correct to 3 significant figures) 2
- (e) After an 18% increase the price of a TV is \$1200.
What was the original price before the increase? 1
- (f) Express $0.\dot{3}0\dot{3}$ as a fraction in lowest terms. 1
- (g) Prove $\sin^2 \theta \cos^2 \theta + \sin^4 \theta = \sin^2 \theta$. 2
- (h) Find a if $\sqrt{a} = \sqrt{20} + \sqrt{125}$ 2

END Q1

Question 2(Start a New Booklet) (12 Marks)

(a)



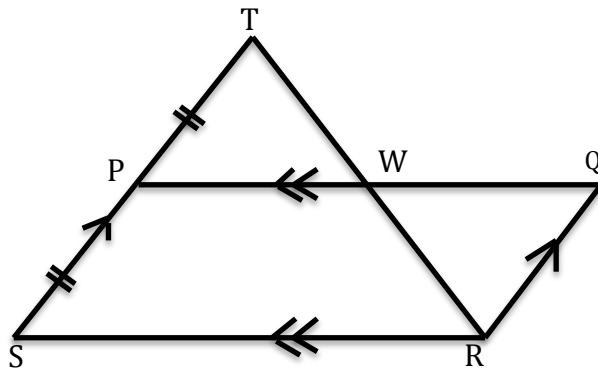
- (i) Find the size of θ to the nearest minute. 2
- (ii) Find the area of $\triangle ABC$ to the nearest m^2 . 2
- (b)
- If $\frac{d^2y}{dx^2} = 6x - 4$ and when $x = 1$, $\frac{dy}{dx} = 7$ and $y = 12$ respectively. 2
- Find y in terms of x .

- (c) Evaluate $\sum_{r=2}^6 12 - 2r$ 2
- (d) Differentiate with respect to x :
- (i) $\tan 3x$ 1
- (ii) $(2x+1)^5$ 1
- (iii) $\log_e (\sin x)$ 2

END Q2

Question 3(Start a New Booklet) (12 Marks)

(a)



In the above diagram, P is the midpoint of TS, PQ is parallel to SR and TS is parallel to QR.

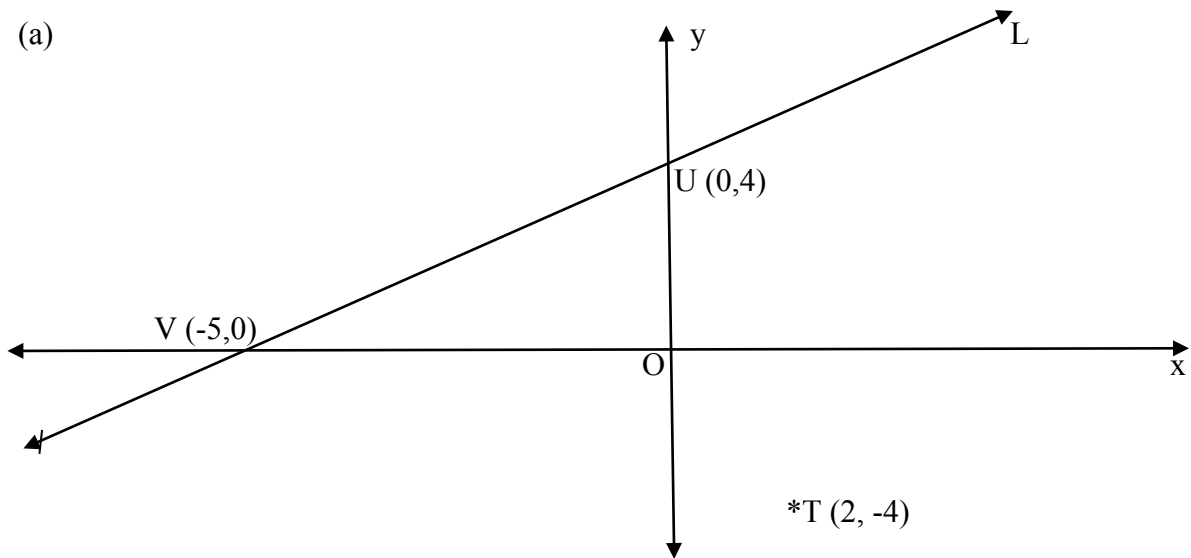
- (i) Prove $\triangle TPW \equiv \triangle WQR$ 2
- (ii) Hence state why W is the midpoint of PQ. 1
- (b) Find $\lim_{x \rightarrow 2} \frac{x-3}{x^2-9}$ 2
- (c) Evaluate $\int_0^4 \frac{9x^2}{3+x^3} dx$ correct to 2 decimal places. 3
- (d) For the equation $2x^2 + x - 3 = 0$ with roots α and β find the value of
- (i) $\alpha\beta$ 1
- (ii) $\alpha + \beta$ 1
- (iii) $\alpha^2 + \beta^2$ 2

END Q3

Question 4(Start a New Booklet)

(12 Marks)

(a)



The line L crosses the x axis at V (-5, 0) and the y axis at U (0,4). The point T (2, -4) is also shown. O is the origin.

- | | | |
|------|--|---|
| (i) | Find the gradient of the line L. | 1 |
| (ii) | Show the equation of the line L is $4x - 5y + 20 = 0$ | 2 |
| (i) | Find the perpendicular distance from the point T(2,-4) to the line L.
(Leaving your answer in exact form) | 2 |
| (iv) | Find the distance VU. (Leaving your answer in exact form) | 1 |
| (v) | Find the area of triangle VUT. | 1 |
| (vi) | At what angle is the line L inclined to the positive x axis?
(Answer to the nearest minute) | 2 |
| (b) | Find A, B and C if $2x^2 - 5x + 7 \equiv 2A(x + 1)^2 + B(x + 1) + C$ | 3 |

END Q4

Question 5(Start a New Booklet)

(12 Marks)

(a) For the parabola with equation $x^2 = 25y$ at the point (5,1) on it find:

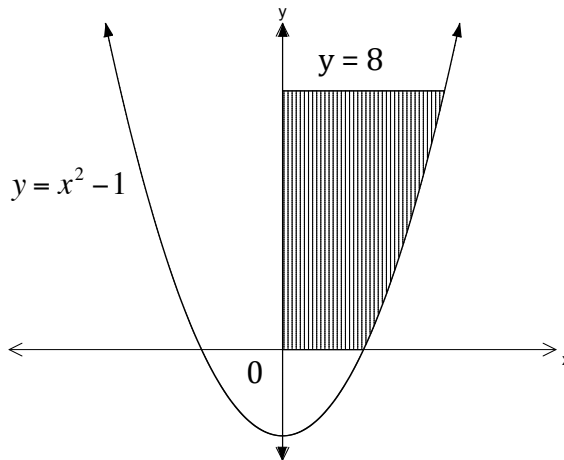
- | | | |
|------|-----------------------------|---|
| (i) | The gradient of the tangent | 1 |
| (ii) | The equation of the tangent | 2 |

- (b) For the curve $y = x^3 - 3x^2 - 9x + 15$
- (i) Find the y intercept. 1
 - (ii) Find the turning points determining their nature. 4
 - (iii) Find and test the point of inflexion. 2
 - (iv) Sketch the curve clearly showing the above features. 2

END Q5

Question 6(Start a New Booklet) (12 Marks)

- (a) Prove that $\frac{\cos \theta}{1 - \sin \theta} - \frac{\cos \theta}{1 + \sin \theta} = 2 \tan \theta$ 3
- (b) The diagram shows the region bounded by the curve $y = x^2 - 1$, the line $y = 8$ and the x and y axes.



Find the volume of the solid of revolution formed when the region is rotated about the y axis. (Leave your answer in exact form) 3

- (c) Use Simpson's rule with 3 function values to find a 2 decimal place approximation to $\int_1^7 \sqrt{x} \, dx$. 3
- (d) Find $\int (4x + 1)^2 \, dx$ 3

END Q6

Question 7(Start a New Booklet)

(12 Marks)

- (a) Nic is training for a local marathon.
He has trained by completing practice runs over the marathon course.
So far he has completed three practice runs with times shown below.



Week 1	Week 2	Week 3
3 hours	2 hours 51 minutes	2 hours 42 minutes 27 seconds

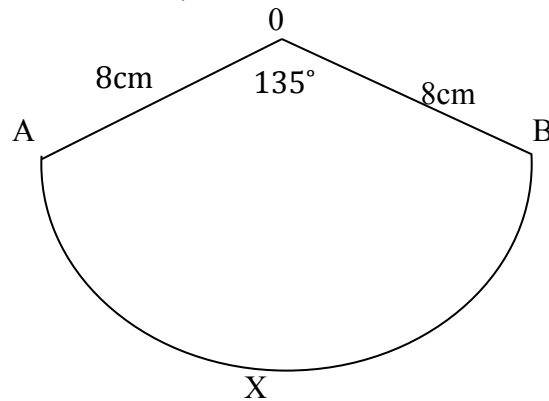
- (i) Show that these times form a geometric series with a common ratio $r = 0.95$. 1
- (ii) If this series continues, what would be his expected time in Week 5, to the nearest second? 1
- (iii) How many hours, minutes and seconds (to the nearest second) will he have run in total in his practice runs in these 5 weeks? 2
- (b) For the parabola with equation $x^2 + 4 = 4y$ find the;
- (i) Vertex 1
- (ii) Focal Point 1
- (c) Given that the limiting sum of the series $1 - 2x + 4x^2 - 8x^3 + \dots$ is $\frac{3}{5}$, find x . 2
- (d) In radioactive elements, the rate of decay is proportional to the mass present given by the formulae $M = M_0 e^{kt}$.
- (i) If it takes 300 years for the mass of a piece of radium to decrease from 10grams to 6 grams, find the value of k to 5 decimal places. 2
- (ii) Find the half-life of the material, to the nearest year.
(Half-life is the time taken for the element to lose half its mass) 2

END Q7

Question 8(Start a New Booklet)

(12 Marks)

(a)

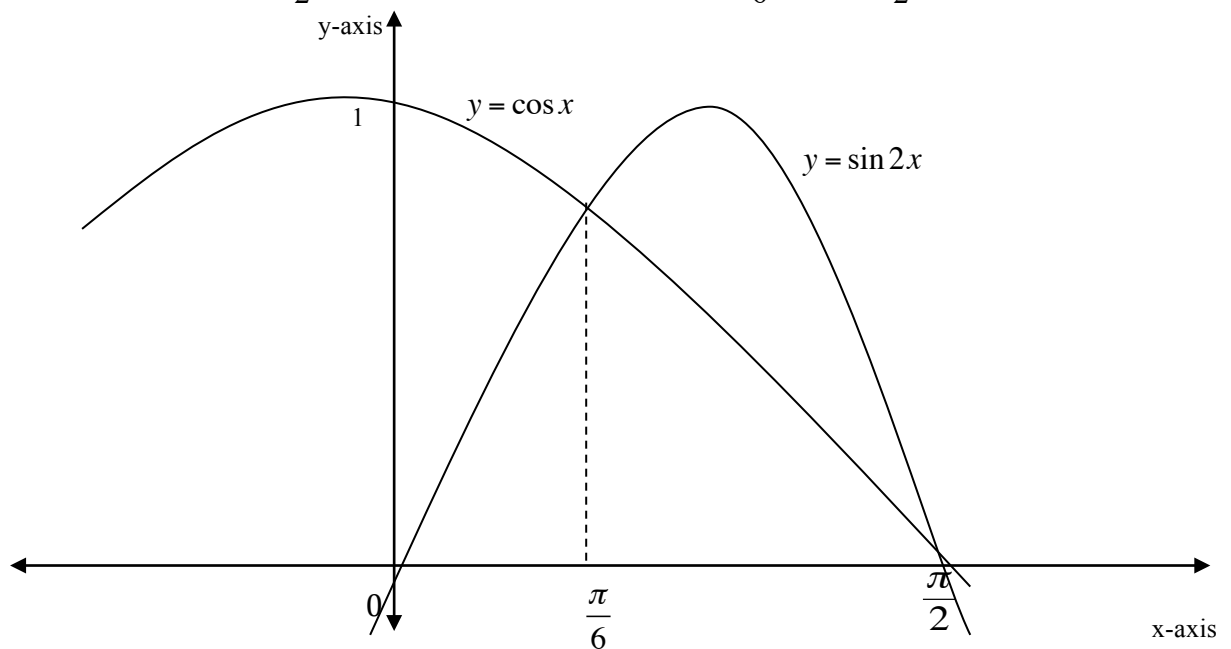


A piece of paper is in the shape of a sector of a circle. The radius is 8cm and the angle at the centre is 135° . The straight edges of the sector are placed together so that a cone is formed.

(i) Find the exact value of the arc length AXB. 1

(ii) Find the length of radius of the base of the cone. 2

(b) The diagram below shows the graphs of $y = \sin 2x$ and $y = \cos x$ between $x = 0$ and $x = \frac{\pi}{2}$. The two graphs intersect at $x = \frac{\pi}{6}$ and $x = \frac{\pi}{2}$.



Calculate the area of the region enclosed by the x-axis and the curves $y = \cos x$ and $y = \sin 2x$.

4

- (c) A dad has 5 tickets in a footy raffle in which there are two separate prizes to be won and 100 tickets are sold.
- (i) Draw a tree diagram representing the outcomes of this raffle. 1
 - (ii) Find the probability that the dad wins only second prize. 2
 - (iii) Find the probability that the dad wins both prizes. 2

END Q8

Question 9(Start a New Booklet) (12 Marks)

- (a) A car dealership has a car for sale for a cash price of \$20 000. It can also be bought on terms over three years. The first six months are interest free and after that interest is charged at the rate of 1% per month on that months balance. Repayments are to be made in equal monthly installments beginning at the end of the first month.

A customer buys the car on these terms and agrees to monthly repayments of \$ M . Let \$ A_n be the amount owing at the end of the n th month.

- (i) Find an expression for A_6 . 1
 - (ii) By first finding A_7 , show that :

$$A_8 = (20\,000 - 6M)1.01^2 - M(1 + 1.01)$$
 2
 - (iii) Find an expression for A_{36} . 2
 - (ii) Assuming the car is paid off after 36 months, find the value of M . (Answer correct to the nearest \$) 3
- (b) A particle moves along the x-axis, its distance x cm from the origin being given by $x = 2\sin\frac{1}{3}t$, where t is in seconds.
- (i) Find the velocity equation in terms of t . 1
 - (ii) Sketch the velocity graph of this particle for $0 \leq t \leq 6\pi$. 2
 - (iii) Using your graph or otherwise find the first time the particle is stationary. 1

END Q9

Question 10(Start a New Booklet)

(12 Marks)

- (a) A newly married couple want to build up a deposit to buy their first house. They create a savings plan in which they deposit \$500 on the first day of each month into an account which pays a fixed rate of interest of 6% per annum, compounded monthly. They start this savings plan on the first of January 2009 and hope to take the money out at the end of December 2011.
- (i) How much will be in the account after 1 month? 1
- (ii) Show that the amount they have in the account by the end of February 2009 is $A_2 = 500(1.005^2 + 1.005)$ 2
- (iii) How much will they have in the account at the end of December 2011? 2
- (b) John, Sam and William enter a golf tournament and the probabilities that each will win are $\frac{1}{5}$, $\frac{2}{7}$ and $\frac{3}{10}$ respectively. Assuming there is an outright winner of the tournament, find the chance (in lowest fractional form) that it is one of these three golfers. 2
- (c) A rectangular sheet of metal measures 200cm by 100cm. Four equal squares with side lengths x cm are cut out of all the corners and then the sides of the sheet are turned up to form an open rectangular box.
- (i) Draw a diagram representing this information. 1
- (ii) Show that the volume of the box can be represented by the equation $V = 4x^3 - 600x^2 + 20\,000x$ 1
- (iii) Find the value of x such that the volume of the tool box is a maximum. 3
(Answer to the nearest cm)

END Q10

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1 (a) $\frac{7}{25}$

(b) $e^{-0.6} = 0.549$ (3 dp) ✓

(c) $4 - 5x \leq 3$

$4 - 3 \leq 5x$ ✓

$x \geq \frac{1}{5}$ ✓

(d) $\int_0^2 3e^{2x} dx = \left[\frac{3}{2} e^{2x} \right]_0^2$ ✓

$= \frac{3}{2} (e^4 - e^0)$

$= \frac{3}{2} e^4 - \frac{3}{2}$ ✓

$= 80.4$ (3 sf) ✓

(e) $1.18 \times x = 1200$ $x = \text{original price}$

$x = \frac{1200}{1.18}$

$x = \$1016.95$ ✓

(f) $x = 0.303$

$1000x = 303.303$

$999x = 303$

$x = \frac{303}{999}$

$x = \frac{101}{333}$ ✓

(g) Prove $\sin^2 \theta \cos^2 \theta + \sin^4 \theta = \sin^2 \theta$

$\sin^2 \theta (\cos^2 \theta + \sin^2 \theta) = \sin^2 \theta$

$\sin^2 \theta (1) = \sin^2 \theta$ ✓

$\sin^2 \theta = \sin^2 \theta$ as req ✓

(h) $\sqrt{a} = \sqrt{20} + \sqrt{125}$

$\sqrt{a} = 2\sqrt{5} + 5\sqrt{5}$ ✓

$\sqrt{a} = 7\sqrt{5}$

$\sqrt{a} = \sqrt{49 \times 5}$

$\sqrt{a} = \sqrt{245}$

$\therefore a = 245$ ✓

Q2 (a)

(i) $\frac{\sin \theta}{7.9} = \frac{\sin 82^\circ 18'}{9.6}$ ✓

$\sin \theta = \frac{7.9 \times \sin 82^\circ 18'}{9.6}$

$\theta = 54^\circ 38'$ ✓

(ii)

Area $\triangle ABC = \frac{1}{2} \times 7.9 \times 9.6 (\sin 43^\circ 4')$ ✓

$= 26 \text{ m}^2$ ✓

(b) $\frac{d^2 y}{dx^2} = 6x - 4$

$\frac{dy}{dx}$

$dy = 3x^2 - 4x + C$

$\frac{dx}$

$y = 3 - 4 + C$

$y = -1 + C$ ✓

$C = 8$ $\frac{dy}{dx} = 3x^2 - 4x + 8$

$y = x^3 - 2x^2 + 8x + C$

$12 = 1 - 2 + 8 + C$

$12 = 7 + C$

$C = 5$ ✓

$\therefore y = x^3 - 2x^2 + 8x + 5$ ✓

(c) $\sum_{r=2}^6 12 - 2r = 8 + 6 + 4 + 2 + 0$ ✓

$= 20$ ✓

(d) (i) $\frac{d}{dx} \tan 3x = 3 \sec^2 3x$

(ii) $\frac{d}{dx} (2x+1)^5 = 5 \times 2 (2x+1)^4$ ✓

$= 10(2x+1)^4$ ✓

(iii) $\frac{d}{dx} \log_e (\sin x) = \frac{\cos x}{\sin x}$ ✓

$= \cot x$ ✓

1 }
2 } OS
3 }
4 } PR
5 }
6 }
7 } PA
8 }
9 } T.G.
10 }

Q3 (a) (i) $\angle TWP = \angle QWR$ (vert opp)

$\angle PTR = \angle QRW$ (alt $\angle$ between 11 lines)

$PQRS$ is a parallelogram. $PQ \parallel SR$, $PS \parallel QR$

$PS = QR$ (opp sides of parallelogram)

P is given Midpoint, $\therefore PS = QR = PT$ hence by AAS $\triangle TPW \cong \triangle QWR$

(ii) By corresponding sides of congruent triangles $PW = WR$ hence W is midpoint of PQ

(iv) perp dist = $|4(2) - 5(-4) + 20| \sqrt{4^2 + (-5)^2}$

$$= \frac{|8 + 20 + 20|}{\sqrt{16 + 25}}$$

$$= \frac{48}{\sqrt{41}}$$

$$= \frac{48}{\sqrt{41}}$$

(iv) dist $UV = \sqrt{(-5-0)^2 + (0-4)^2}$

$$= \sqrt{25 + 16}$$

$$= \sqrt{41}$$

(v) Area $\triangle UUT = \frac{1}{2} \times \sqrt{41} \times \frac{48}{\sqrt{41}}$

$$= 24 \text{ u}^2$$

(vi) $\tan \theta = \frac{4}{5}$

$$\theta = \tan^{-1}\left(\frac{4}{5}\right)$$

$$\therefore \theta = 38^\circ 40'$$

(b) $\lim_{x \rightarrow 2} \frac{x-3}{x^2-9} = \lim_{x \rightarrow 2} \frac{(x-3)}{(x-3)(x+3)}$

$$= \frac{1}{5}$$

(c) $\int_0^4 \frac{9x^2}{3+x^3} = \left[3 \ln(3+x^3) \right]_0^4$

$$= 3 (\ln 7 - \ln 3)$$

$$= 9.32$$

(d) $2x^2 + x - 3 = 0$

(i) $\alpha\beta = -3$

(ii) $\alpha + \beta = -\frac{1}{2}$

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= \left(-\frac{1}{2}\right)^2 - 2\left(-\frac{3}{2}\right)$$

$$= \frac{1}{4} + 3$$

$$= 3\frac{1}{4}$$

Q4 (a) (i) $m_L = \frac{4-0}{0-5} = -\frac{4}{5}$

(ii) $4 - 4 = -\frac{4}{5}(x - 0)$

$$54 - 20 = 4(x - 0)$$

$$54 - 4x - 20 = 0$$

$$4x - 54 + 20 = 0 \text{ as req.}$$

(b)

$$2x^2 - 5x + 7 = 2A(x+1)^2 + B(x+1) + C$$

$$A = 1$$

$$2x^2 - 5x + 7 = 2(x+1)^2 + B(x+1) + C$$

Let $x = -1$

$$2 + 5 + 7 = 0 + 0 + C$$

$$\therefore C = 14$$

$$2x^2 - 5x + 7 = 2(x+1)^2 + B(x+1) + C$$

Let $x = 0$

$$7 = 2 + B + 14$$

$$7 - 16 = B$$

$$B = -9$$

Hence $A = 1$, $B = -9$, $C = 14$

Q2/ (a) $x = 25y$

$y = \frac{x^2}{25}$

(i) $\frac{dy}{dx} = \frac{2}{25}x$

$m_t = \frac{2}{25}(5)$

$m_t = \frac{10}{25}$

$m_t = \frac{2}{5}$

(ii) $y-1 = \frac{2}{5}(x-5)$

$y-1 = \frac{2}{5}x - 2$

$y = \frac{2}{5}x - 1$

OR

$5y = 2x - 5$

$2x - 5y - 5 = 0$

(b) $y = x - 3x^2 - 9x + 15$

(i) y -intercept is 15

(ii) $y' = 3x^2 - 6x - 9$

$y'' = 6x - 6$

let $y' = 0$ for stat pts

$3x^2 - 6x - 9 = 0$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0$

$x = 3, x = -1$

den $x = 3, y = 27 - 27 - 27 + 15$

$y = -12$

den $x = -1, y = -1 - 3 + 9 + 15$

$y = 20$

∴ Stat pts $(3, -12), (-1, 20)$

Testing when $x = 3$

$y'' = 18 - 6$

$y'' = 12 \uparrow \uparrow$ concave up

∴ Min TP at $(3, -12)$

Testing when $x = -1$

$y'' = -6 - 6$

$y'' = -12 \downarrow \downarrow$

concave down

∴ Max TP at $(-1, 20)$

(iii) let $6x - 6 = 0$ (for pts of inflection)

$6x = 6$

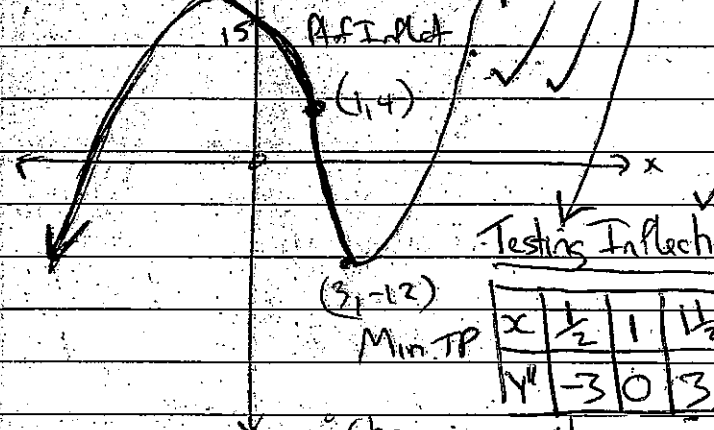
$x = 1$

$y = 1 - 3 - 9 + 15$

$y = 4$

∴ Pt of inflection at $(1, 4)$

Max TP $(-1, 20)$



Testing Inflection

x	1/2	1	3/2
y''	-3	0	3

∴ Change in concavity

Q6/ (a) Prove $\frac{\cos \theta}{1 - \sin \theta} - \frac{\cos \theta}{1 + \sin \theta} = 2 \tan \theta$

$\frac{\cos \theta (1 + \sin \theta) - \cos \theta (1 - \sin \theta)}{1 - \sin^2 \theta} = 2 \tan \theta$

$\frac{\cos \theta + \cos \theta \sin \theta - \cos \theta + \cos \theta \sin \theta}{\cos^2 \theta} = 2 \tan \theta$

$\frac{2 \cos \theta \sin \theta}{\cos^2 \theta} = 2 \tan \theta$

$\frac{2 \sin \theta}{\cos \theta} = 2 \tan \theta$

∴ $2 \tan \theta = 2 \tan \theta$ as req.

P 3/6

(b) $y = x^2 - 1$
 $y + 1 = x^2$
 $V = \pi \int_0^8 y + 1 \, dy$
 $= \pi \left[\frac{y^2}{2} + y \right]_0^8$
 $= \pi [(32 + 8) - 0]$
 $= 40\pi \, \text{m}^3$

(c)

x	1	4	7
y	1	2	$\sqrt{7}$

 $h = 7 - 1 = 6$
 $h = 3$
 $\int_1^7 \sqrt{x} \, dx = \frac{3}{3} (1 + \sqrt{7} + 4(2))$
 $= 1(1 + \sqrt{7} + 8)$
 $= 9 + \sqrt{7}$
 $= 11.65$

(d) $\int (4x+1)^2 \, dx = \frac{(4x+1)^3}{12} + C$
 must have C to gain 2 marks.

Q7 (i) $2.85 = 0.95$
 $2.7075 = 0.95$
 2.85
 OR similar method.

(ii) $T_5 = 2.7075 \times 0.95 \times 0.95$
 $= 2 \text{ hr } 26 \text{ min } 37 \text{ sec}$

(iii) $S_5 = \frac{3(1 - 0.95^5)}{1 - 0.95}$
 $= 13 \text{ hr } 34 \text{ min } 23 \text{ sec}$

(b) $x^2 + 4 = 4y$
 $x^2 = 4y - 4$
 $x^2 = 4(y - 1)$
 $x^2 = 4(\frac{1}{4})(y - 1)$
 (i) Vertex (0, 1)
 (ii) Focal pt (0, $1\frac{1}{4}$)

(e) $\frac{a}{1-r} = \frac{3}{5}$

$a = 1$

$r = -2x$

$\frac{1}{1 - -2x} = \frac{3}{5}$

$\frac{1}{1 + 2x} = \frac{3}{5}$

$5 = 3 + 6x$

$2 = 6x$

$x = \frac{1}{3}$

(d) $M = M_0 e^{kt}$

(i) $6 = 10e^{300k}$

$0.6 = e^{300k}$

$300k = \ln 0.6$

$k = \frac{\ln 0.6}{300} = -0.00170$

(ii) $1 = 2e^{-0.00170t}$

$-0.00170t$

$0.5 = e^{-0.00170t}$

$-0.00170t = \ln 0.5$

$t = \frac{\ln 0.5}{-0.00170}$

$t = 407.7$

$t = 408 \text{ years}$

8. (a) 628

(a)

(i)

$$\text{Arc Length } A \times B = 8 \times \frac{3\pi}{4}$$

(ii)

$$= 6\pi \text{ cm. } \checkmark$$

$$C = 2\pi r$$

$$6\pi = 2\pi r \quad \checkmark$$

$$\frac{6\pi}{2\pi} = r$$

$$r = 3 \text{ cm} \quad \checkmark$$

(b)

$$A = \int_0^{\frac{\pi}{6}} \sin 2x \, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos x \, dx$$

$$= \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}} + \left[\sin x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \quad \checkmark$$

$$= -\frac{1}{2} \left[\cos \frac{\pi}{3} - \cos 0 \right] + \left[\sin \frac{\pi}{2} - \sin \frac{\pi}{6} \right]$$

$$= -\frac{1}{2} \left[\frac{1}{2} - 1 \right] + \left[1 - \frac{1}{2} \right]$$

$$= \frac{1}{4} + \frac{1}{2}$$

$$= \frac{3}{4} \quad \checkmark$$

(c)

$$(i) P(\text{Only win second}) = \frac{95}{100} \times \frac{5}{99} \quad \checkmark$$

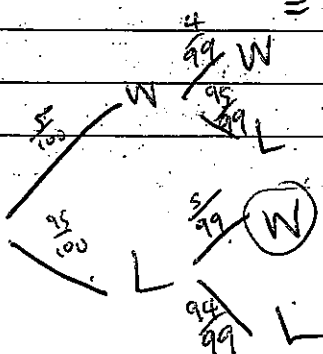
$$= \frac{19}{396}$$

$$= 0.0479 \quad \checkmark$$

$$(ii) P(\text{Win both prizes}) = \frac{5}{100} \times \frac{4}{99} \quad \checkmark$$

$$= \frac{1}{495}$$

(i)



$$(a) (i) A_6 = 20000 - 6M \quad \checkmark$$

$$(ii) A_7 = (20000 - 6M)1.01 - M \quad \checkmark$$

$$A_8 = [(20000 - 6M)1.01 - M]1.01 - M \quad \checkmark$$

$$= (20000 - 6M)(1.01)^2 - 1.01M - M$$

$$\therefore A_8 = (20000 - 6M)(1.01)^2 - M(1 + 1.01) \quad \checkmark$$

(iii)

$$A_{36} = (20000 - 6M)1.01^{30} - M(1 + 1.01 + \dots + 1.01^{29}) \quad \checkmark \quad \checkmark$$

(iv)

$$A_{36} = 0 \quad \checkmark$$

$$0 = (20000 - 6M)1.01^{30} - M \frac{(1.01^{30} - 1)}{(1.01 - 1)}$$

$$\frac{M(1.01^{30} - 1)}{0.01} = 20000(1.01^{30}) - 6M(1.01^{30})$$

$$\frac{M(1.01^{30} - 1)}{0.01} + 6M(1.01^{30}) = 20000(1.01^{30})$$

$$M = \frac{20000(1.01^{30})}{\frac{1.01^{30} - 1}{0.01} + 6(1.01^{30})} \quad \checkmark$$

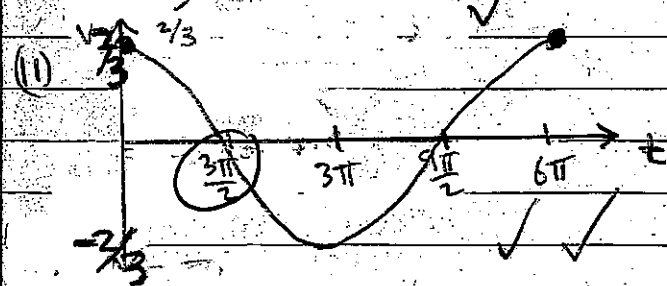
$$M = \$628.77$$

$$M = \$629 \text{ (nearest \$)} \quad \checkmark$$

$$(b) x = 2 \sin \frac{1}{3} t$$

$$v = x' = \frac{2}{3} \cos \frac{1}{3} t$$

$$(i) v = \frac{2}{3} \cos \frac{1}{3} t$$



(iii) Particle is 1st stationary

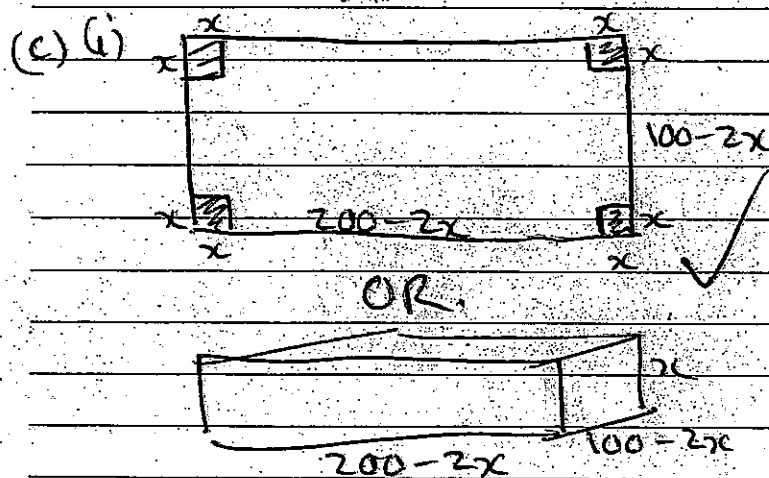
$$v = 0, \text{ when } t = \frac{3\pi}{2} \text{ sec.} \quad \checkmark \quad \text{PS 16}$$

$r = \frac{0.06}{12} = 0.005$
 (a) (i) $A_1 = 500(1.005)$
 $= \$502.50$

(ii) $A_2 = 500(1.005)(1.005)$
 $+ 500(1.005)$
 $= 500(1.005)^2 + 500(1.005)$
 $= 500(1.005^2 + 1.005)$ ✓
 as req ✓

(iii) $A_{36} = 500(1.005^{36} + 1.005^{35} + \dots$
 $\dots + 1.005^1)$ ✓
 $= 500(1.005)(1.005^{36} - 1)$
 $\quad \quad \quad 0.005$
 $A_{36} = \$19\,766.39$ ✓

(b) $P(\text{E. New, John, Sam or William Wins})$
 $= \frac{1}{5} + \frac{2}{7} + \frac{3}{10}$
 $= \frac{11}{14}$



(ii) $V = x(100-2x)(200-2x)$

$V = 4x^3 - 600x^2 + 20000x$ ✓

(iii) $V' = 12x^2 - 1200x + 20000$

Let $V' = 0$ for stat pts.

$6x^2 - 600x + 10000 = 0$

$x = 600 \pm \sqrt{600^2 - 4 \times 6 \times 10000}$

$x = 78.87, x = 21.13$

$V'' = 24x - 1200$

Testing

When $x = 78.87$

$V'' = 692 \therefore$ Concave up ✓
 Min TP at $x = 78.8$

When $x = 21.13$

$V'' = -692 \therefore$ Concave down ✓

Max TP at $x = 21.13$

$\therefore$ Max Vol achieved ✓

When $x = 21 \text{ cm}$